MOLECULAR INTERACTION ANALYSIS

Surface Residue Selection and Mutagenesis Rationale — Helix H1

Protein: 1R5E | Helix H1: SSQLLSVRHQLAES | Residues 33-46

1. Summary

This report presents the SASA-based surface residue selection, complete molecular interaction analysis, and mutagenesis rationale for **Helix H1 (residues 33-46, SSQLLSVRHQLAES)** of protein 1R5E. Ten surface-exposed residues were selected as mutation targets. The two buried leucines (**LEU36 and LEU43**) were excluded to preserve structural integrity. A total of **32 molecular interactions** were identified, including 10 inter-helix contacts that anchor H1 to neighbouring structural elements.

2. Background and Rationale

Mutagenesis of helical residues is most informative when restricted to surface-exposed positions for three key reasons:

- **Structural safety**: Buried hydrophobic residues (LEU36 SASA=3.09 Angstrom², LEU43 SASA=0.00 Angstrom²) contribute to hydrophobic core packing. Mutating them risks helix destabilisation or global unfolding.
- **Functional relevance**: Surface residues mediate protein-protein interactions, electrostatic contacts, and solvent-dependent chemistry.
- **Predictable outcomes**: Surface mutations have lower energetic costs, enabling rational engineering without thermodynamic destabilisation.

3. SASA-Based Residue Classification

A Solvent Accessible Surface Area (SASA) threshold of **approximately 25 Angstrom²** was applied. Surface residues (SASA > 25 Angstrom²) in confirmed helical secondary structure were retained as mutation targets. The 10 selected surface residues span 40.68 to 138.35 Angstrom².

Pos	Amino Acid	SASA	Classification
33	SER	65.80	Surface
34	SER	79.78	Surface
35	GLN	68.14	Surface
36	LEU	3.09	Buried
37	LEU	40.68	Surface
38	SER	53.48	Surface
39	VAL	16.33	Partial
40	ARG	68.05	Surface
41	HIS	83.74	Surface

Pos	Amino Acid	SASA	Classification
42	GLN	74.40	Surface
43	LEU	0.00	Buried
44	ALA	27.41	Partial
45	GLU	138.35	Surface
46	SER	59.92	Surface

Green: Surface targets **Red:** Buried/excluded **Yellow:** Partially exposed

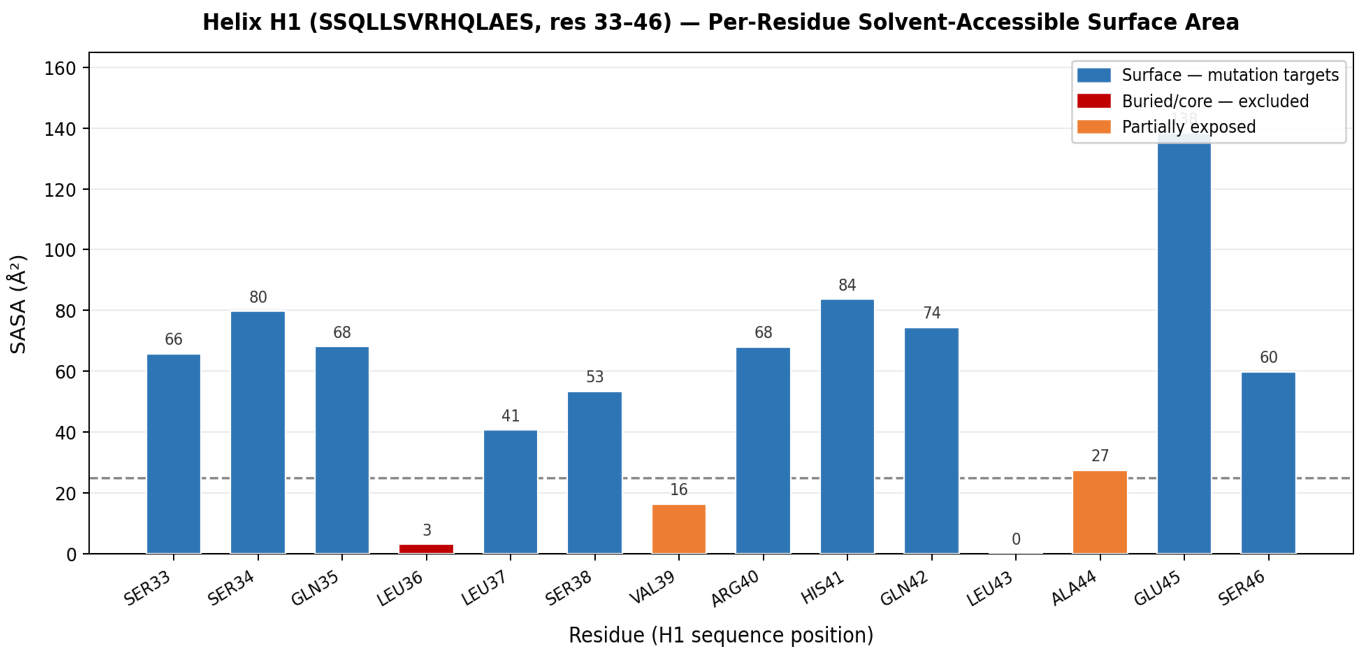

Figure 1. Per-residue SASA for Helix H1 (residues 33-46, SSQLLSVRHQLAES). Blue = surface mutation targets; red = buried/core (excluded); orange = partially exposed. Dashed line = 25 Angstrom² classification threshold.

4. Excluded Core Residues

Residue	SASA	Class	Structural Rationale for Exclusion from Mutation
LEU36	3.09	Buried	Very low SASA. Forms alkyl contacts with VAL31, PRO102, and HIS87 (Pi-Alkyl). Provides core hydrophobic packing essential for helix stability. Backbone H-bond donor interactions present but structural only.
LEU43	0.00	Buried	Zero SASA which means it is fully inaccessible. Forms alkyl contacts with LEU80, MET109, and ALA112. Backbone H-bond acceptor for SER46. Its complete burial and multiple hydrophobic contacts make it an essential structural residue.

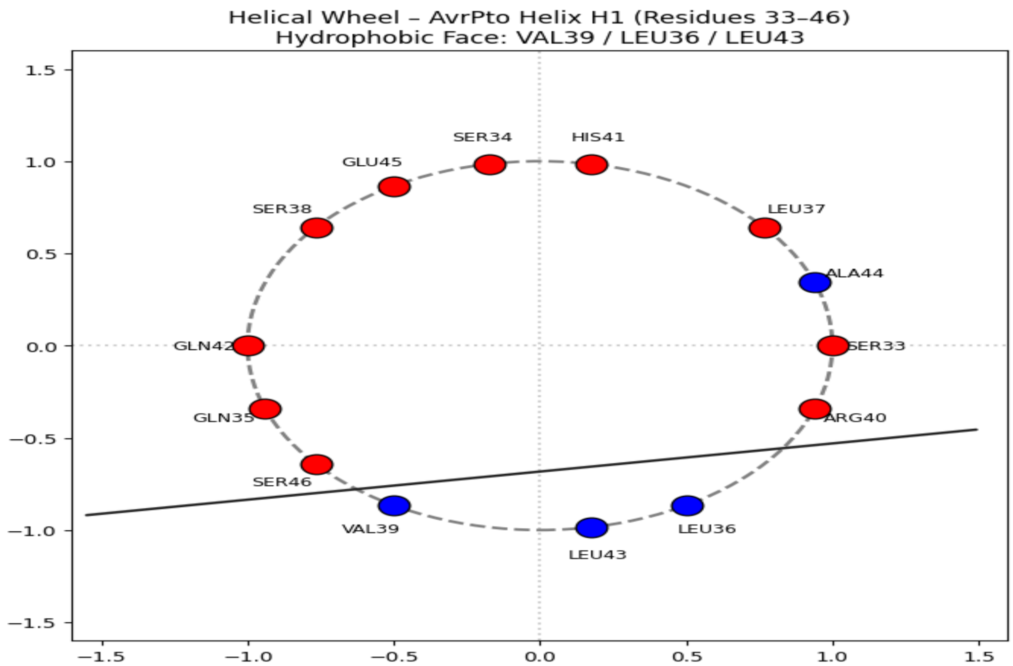

Figure 2. Helical wheel projection of the peptide SSQLLSVRHQLAES, showing residue distribution around the α-helix axis and highlighting amphipathic character.

5. Molecular Interactions of Surface Residues using BIOVIA

The 32 filtered interactions for the **10 surface residues** are categorised below. Intra-H1 interactions stabilise the helix backbone; inter-helix interactions are key functional contacts with neighbouring structural elements.

Residue	SASA	Interaction Types	Interaction Details	Zone
SER33	65.80	H-bond (backbone acceptor)	H-bond from LEU36 and LEU37 backbone NH; inter-helix H-bond from HIS87 side chain NE2	**Intra + Inter**
SER34	79.78	Conventional H-bond	Backbone H-bond acceptor for SER38; donor to THR32 OG1 (N-cap); C-H bond from SER38	**Intra + Inter**
GLN35	68.14	H-bond + Carbon H-bond	H-bond donor to THR32 backbone O; intramolecular side-chain H-bond (HE21-O); C-H bonds from SER38	**Intra + Inter**
LEU37	40.68	Pi-Alkyl + H-bond	HIS41 pi-alkyl; HIS87 pi-alkyl (inter-helix); backbone acceptor for HIS41 H and ARG40 HD2/HD3	**Intra + Inter**
SER38	53.48	H-bond + Carbon H-bond	Donor to SER34 backbone O; acceptor from GLN42 H; C-H bonds to SER34 O and GLN35 O	**Intra**
ARG40	68.05	Salt bridge + Electrostatic + H-bond + Alkyl	3 contacts to ASP84 (salt bridge + H-bond + electrostatic); Pi-cation to HIS41; alkyl to LEU80 and ALA83	**Intra + Inter**
HIS41	83.74	Electrostatic + Pi-Alkyl + H-bond	Electrostatic to ASP84; pi-alkyl to LEU37; backbone H-bond; C-H bond to LEU37 O	**Intra + Inter**
GLN42	74.40	H-bond + Carbon H-bond	Donor to SER38 backbone O; carbon H-bond to GLU45 OE1	**Intra**
GLU45	138.35	Conventional H-bond	H-bond donor to HIS41 backbone O; C-H bond acceptor from GLN42; highest SASA in helix	**Intra**
SER46	59.92	H-bond + Carbon H-bond	Donor to GLN42 and LEU43 backbone O; C-H bond to LEU43 O; inter-helix acceptor from GLY48	**Intra + Inter**

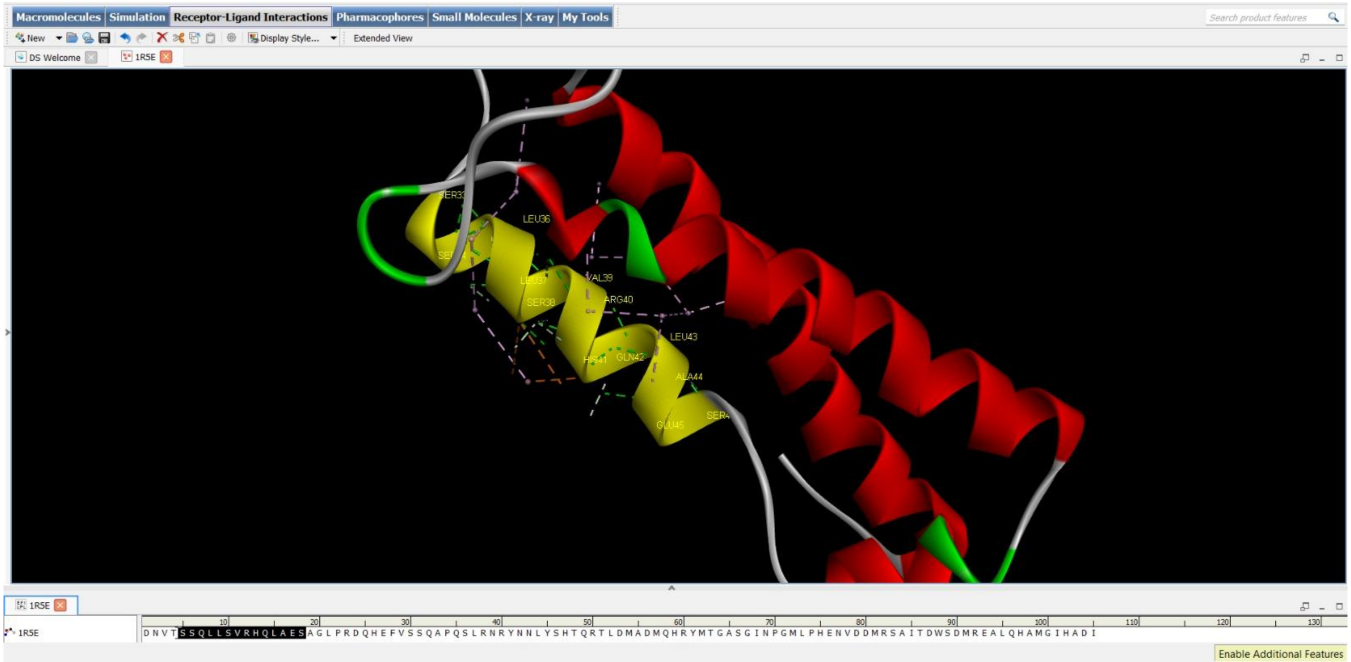

Figure 3. Structural visualization of AvrPto protein (PDB ID: 1R5E) in BIOVIA Discovery Studio. The amino acid sequence is shown below, with residues of helix H1 highlighted to indicate the region analyzed (Yellow color coded) for solvent accessibility and interaction mapping

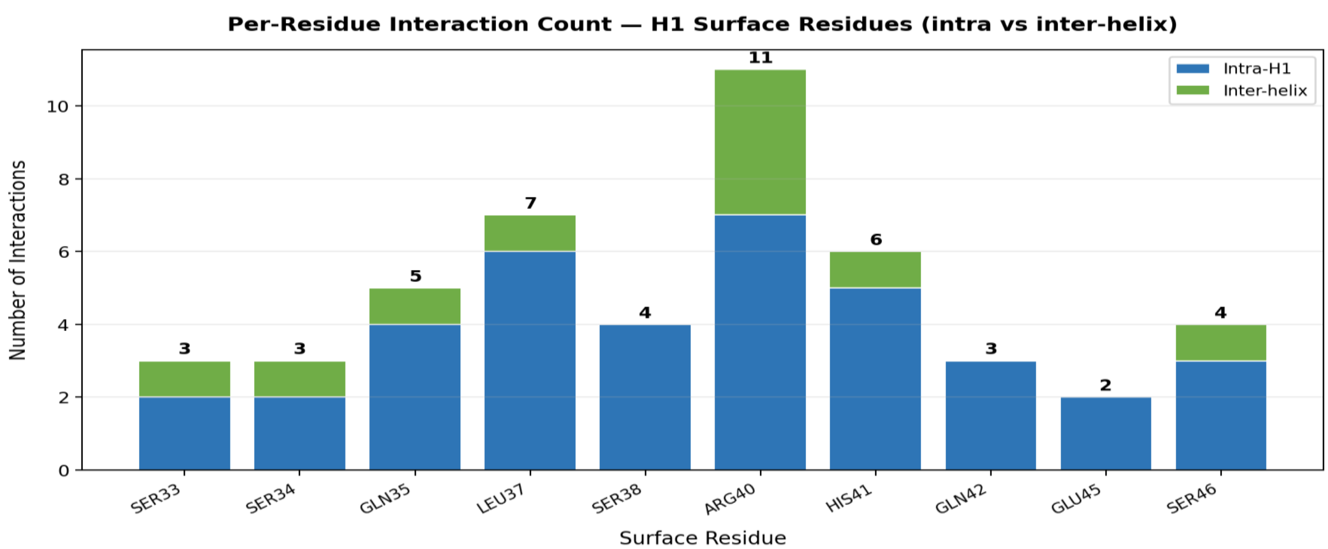

Figure 4. Interaction count per surface residue split by intra-H1 (blue) and inter-helix (green). ARG40 is the dominant inter-helix interaction hub with 7 total contacts.

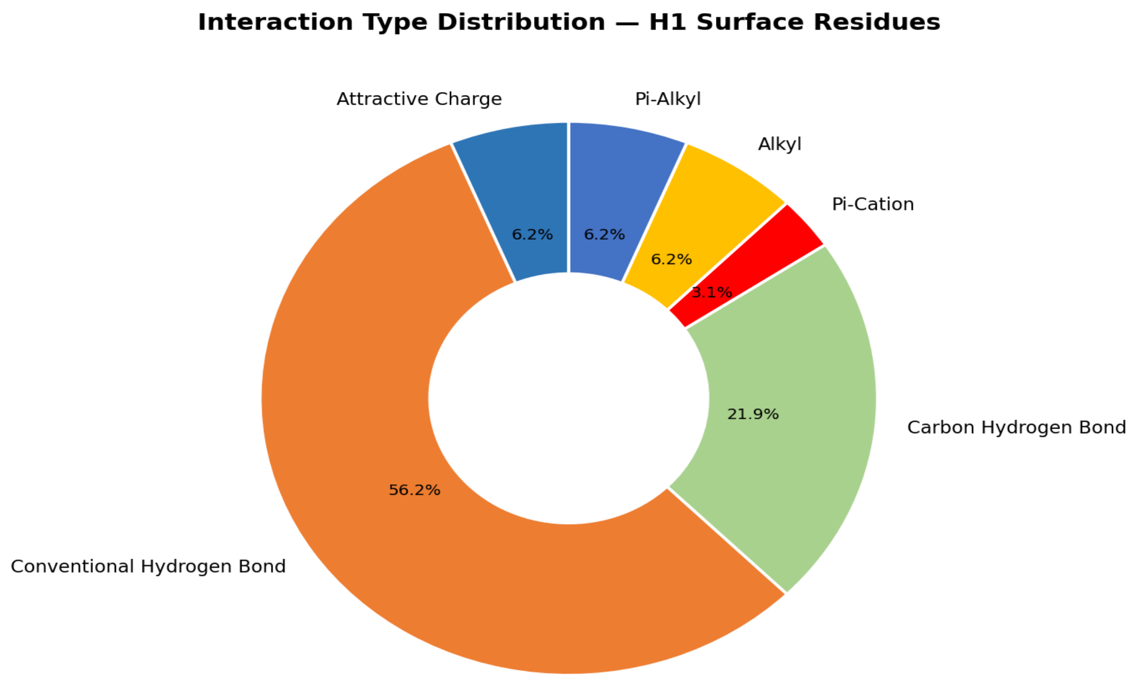

Figure 5. Distribution of interaction types across all 32 surface residue interactions. Conventional H-bonds dominate (helix backbone relay); salt bridge and electrostatic interactions are concentrated at ARG40 and HIS41.

5.2 Key Interactions — Detailed Notes

ARG40 — Inter-Helix Electrostatic Hub

ARG40 forms the richest interaction profile of any H1 residue with 7 interactions total: a salt bridge and two H-bonds to ASP84 (the primary inter-helix anchor), a Pi-cation interaction with HIS41 within the helix, and alkyl contacts with LEU80 and ALA83. This makes ARG40 the central inter-helix tether of H1. Arg->Ala mutation will simultaneously ablate all electrostatic and hydrophobic contacts to neighbouring helices.

HIS41 — pH-Sensitive Functional Residue

HIS41 has the highest SASA (83.74 Angstrom²) among non-glutamate residues. Its imidazole ring participates in an electrostatic attraction to ASP84 and pi-alkyl contact with LEU37, while the backbone NH accepts an H-bond from GLU45. Its pH-dependent protonation (pKa approximately 6.5) means it can toggle charge at physiological pH. His->Ala removes the imidazole entirely; His->Asn retains polarity without charge.

GLU45 — Highest SASA, Electrostatic Surface Anchor

GLU45 carries the largest SASA in the entire helix (138.35 Angstrom²). It contributes an H-bond to HIS41 backbone O and serves as C-H bond acceptor from GLN42. Its primary role is defining the local electrostatic surface potential at the helix C-terminus. Glu->Gln neutralises charge with minimal steric change; Glu->Ala provides complete side-chain ablation.

SER33, SER34, SER38, SER46 — Serine Cluster

Four serine residues span H1 (SASA 53.48-79.78 Angstrom²). SER33 is the target of an inter-helix H-bond from HIS87 NE2, making it a functional N-terminal contact. SER34 donates an H-bond to THR32 OG1 (N-cap). SER38 and SER46 participate in the backbone H-bond relay within the helix.

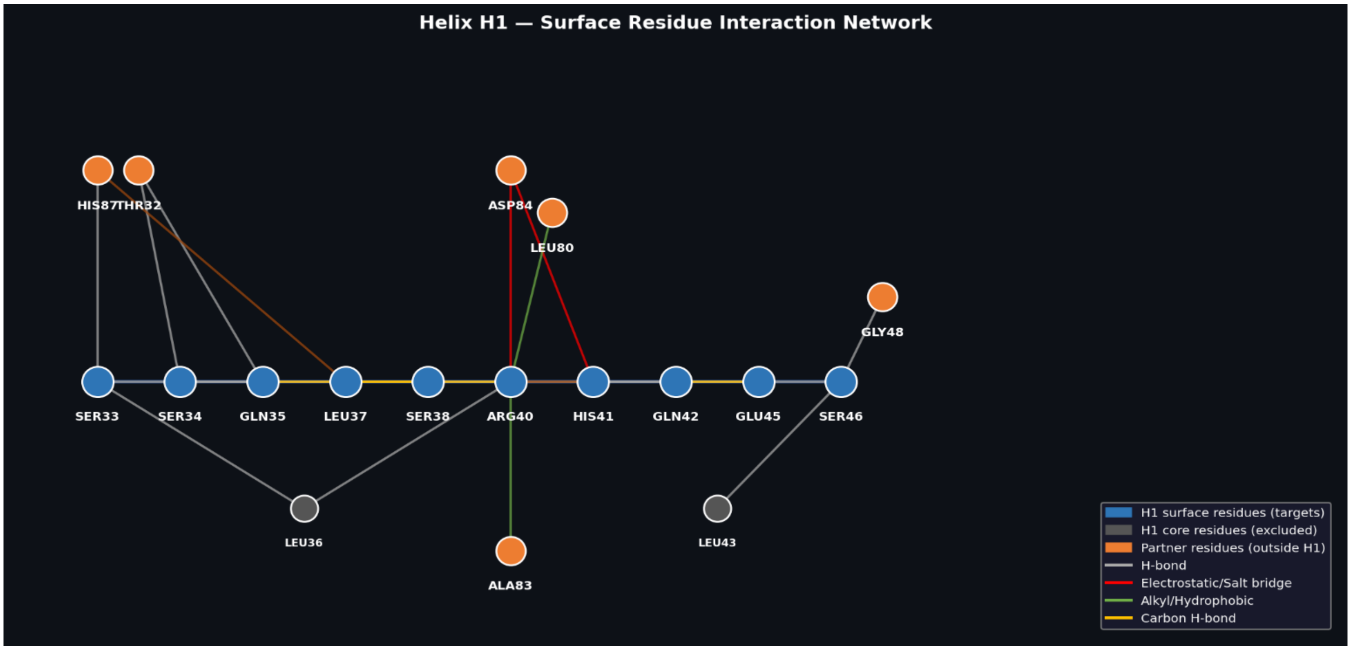

Figure 6. Interaction network for H1 surface residues. Blue nodes = surface targets; grey = excluded core; orange = partner residues outside H1. Grey lines = H-bond; red = electrostatic/salt bridge; green = alkyl/hydrophobic; yellow = carbon H-bond.

Helix 1 Surface Residues and Selection for Mutation Based on Interaction Analysis and Helical Propensity

This report evaluates the ten surface-exposed residues of Helix 1 (SER33–SER46) for suitability as mutation targets. The selection strategy integrates two orthogonal criteria: (i) the nature and criticality of molecular interactions (intrahelical vs. inter-helix), and (ii) α-helix propensity values from the Pace & Scholtz (1996) scale, where lower values denote stronger helix-forming preference (Ala = 0.00 is optimal).

Decision Framework Applied

1. Lower propensity value = more helix-favoring (Ala = 0.00 is optimal; Pro = 3.16 is worst).
2. Residues with Intra-only interactions are free mutation candidates, since no inter-helix contacts to preserve.
3. Residues with critical inter-helix contacts (salt bridges, electrostatics) must be replaced with residues of similar charge; if no better-propensity charged alternative exists, retain the original.
4. N-cap residues (SER34) are structurally essential for helix initiation and **must not be mutated.**
5. If the inter-helix interaction is backbone-mediated (not side-chain dependent), side-chain substitutions are more safely tolerated.
6. SASA values guide solvent exposure: high SASA + Intra-only = highest priority for mutation.

Summary Decision Table

Residue	SASA (Å^2)	Zone	Current Propensity	Decision	Suggested Mutation	New Propensity
SER33	65.80	Intra + Inter	Ser: 0.50	**DO NOT MUTATE**	—	—
SER34	79.78	Intra + Inter	Ser: 0.50	**DO NOT MUTATE**	—	—
GLN35	68.14	Intra + Inter	Gln: 0.39	**CAUTION**	Glu^0 (cautious)	0.16
LEU37	40.68	Intra + Inter	Leu: 0.21	**DO NOT MUTATE**	—	—
SER38	53.48	Intra ONLY	Ser: 0.50	**MUTATE — ALA**	Ala	0.00
ARG40	68.05	Intra + Inter	Arg$^+$: 0.21	**DO NOT MUTATE**	—	—
HIS41	83.74	Intra + Inter	His$^+$: 0.66	**MUTATE — LYS**	Lys	0.26 (vs His$^+$ 0.66)
GLN42	74.40	Intra ONLY	Gln: 0.39	**MUTATE — ALA**	Ala (primary); Glu^0 (secondary)	0.00 (primary); 0.16 (secondary)
GLU45	138.35	Intra ONLY	Glu$^-$: 0.40	**MUTATE — LEU or ALA**	Leu (primary); Ala (secondary)	0.21 (primary); 0.00 (secondary)
SER46	59.92	Intra + Inter	Ser: 0.50	**CONDITIONAL**	Ala (if backbone-mediated contact confirmed)	0.00

Detailed Residue-by-Residue Analysis

SER33 | SASA: 65.80 Å^2 | Intra + Inter | **NOT TO MUTATE**

Interaction Types	H-bond backbone acceptor
Interaction Details	H-bond from LEU36 & LEU37 backbone NH; inter-helix H-bond from HIS87 side chain NE2
Current Propensity	Ser: 0.50
Suggested Mutation	—

Rationale:
The inter-helix H-bond donated by HIS87 side chain NE2 targets SER33, likely engaging the Ser hydroxyl or backbone oxygen. This inter-helix contact is a structural bridge that must be preserved. While Ser has a moderate propensity (0.50), no polar-hydroxyl substitute offers a meaningfully better propensity (Thr: 0.66) to justify the risk of losing the HIS87 inter-helix contact. Mutation to Ala (0.00) would eliminate the H-bond partner entirely. We will retain SER33.

SER34 | SASA: 79.78 Å² | Intra + Inter | NOT TO MUTATE

Interaction Types	Conventional H-bond
Interaction Details	Backbone H-bond acceptor for SER38; donor to THR32 OG1 (N-cap role); C-H bond from SER38
Current Propensity	Ser: 0.50
Suggested Mutation	—

Rationale:
SER34 fulfills a classic N-cap role: its side chain hydroxyl donates to THR32 OG1, capping the helix N-terminus and stabilizing the macrodipole. N-cap positions are among the most structurally sensitive in an α-helix. Serine and Threonine are the canonical N-cap residues; replacing SER34 would destabilize helix initiation. Despite only moderate propensity (0.50), functional importance outweighs the gain from substitution. We will retain SER34.

GLN35 | SASA: 68.14 Å² | Intra + Inter | CAUTION

Interaction Types	H-bond + Carbon H-bond
Interaction Details	H-bond donor to THR32 backbone O; intramolecular side-chain H-bond (HE21–O); C-H bonds from SER38
Current Propensity	Gln: 0.39
Suggested Mutation	Glu⁰ (cautious) (propensity: 0.16)

Rationale:
GLN35 has a decent helix propensity (0.39) and makes an inter-helix H-bond donation to THR32 backbone. The intramolecular side-chain H-bond (HE21–O) suggests partial self-satisfaction. However, the inter-helix donor contact to THR32 means the side-chain participates in cross-helix communication. Mutation to Glu⁰ (propensity 0.16, better) could be explored cautiously, as Glu can also donate H-bonds via its backbone, but the longer side chain alters geometry. We will only consider if the THR32 interaction can be modelled to be preserved. This can be checked in the subsequent tests.

LEU37 | SASA: 40.68 Å² | Intra + Inter | NOT TO MUTATE

Interaction Types	Pi-Alkyl + H-bond
Interaction Details	Pi-alkyl to HIS41; inter-helix pi-alkyl to HIS87; backbone acceptor for HIS41 H and ARG40 HD2/HD3
Current Propensity	Leu: 0.21
Suggested Mutation	—

Rationale:
LEU37 already has excellent helix propensity (0.21, second best after Ala). Its low SASA (40.68 Å²) indicates it is partially buried, and it participates in inter-helix pi-alkyl interaction with HIS87 which is a key hydrophobic contact between helices. Replacing with Ile (0.41) or Met (0.24) would worsen propensity or offer negligible gain, while risking loss of pi-alkyl geometry. Ala (0.00) would eliminate the pi-alkyl contacts. We will retain LEU37.

SER38 | SASA: 53.48 Å² | Intra ONLY | MUTATE — ALA

Interaction Types	H-bond + Carbon H-bond
Interaction Details	Donor to SER34 backbone O; acceptor from GLN42 H; C-H bonds to SER34 O and GLN35 O
Current Propensity	Ser: 0.50
Suggested Mutation	Ala (propensity: 0.00)

Rationale:
SER38 is an Intra-only residue with no inter-helix contacts. All its interactions involve backbone atoms of neighboring residues (SER34 and GLN35), which are mediated primarily through backbone carbonyls and amide NH rather than the Ser side chain. Replacing with ALA (propensity 0.00: best helix former) removes only the side-chain OH while backbone geometry is maintained. This substitution would substantially improve helix propensity with minimal structural disruption. Strong candidate for mutation.

ARG40 | SASA: 68.05 Å² | Intra + Inter | NOT TO MUTATE

Interaction Types	Salt bridge + Electrostatic + H-bond + Alkyl
Interaction Details	3 contacts to ASP84 (salt bridge + H-bond + electrostatic); Pi-cation to HIS41; alkyl contacts to LEU80 and ALA83
Current Propensity	Arg^+: 0.21
Suggested Mutation	—

Rationale:
ARG40 engages in a critical inter-helix salt bridge with ASP84, reinforced by H-bond and electrostatic contacts — three simultaneous contacts to a single partner. To mutate while preserving this interaction, a positively charged residue is required. The only alternative is Lys^+ (propensity 0.26), which is worse than Arg^+ (0.21) in helix propensity. Since no positively charged residue offers better helix propensity than Arg, and the salt bridge network is essential, mutation is not justified. We will retain ARG40.

HIS41 | SASA: 83.74 Å² | Intra + Inter | MUTATE — LYS

Interaction Types	Electrostatic + Pi-Alkyl + H-bond
Interaction Details	Electrostatic interaction to ASP84; pi-alkyl to LEU37; backbone H-bond; C-H bond to LEU37 O
Current Propensity	His^+: 0.66
Suggested Mutation	Lys (propensity: 0.26 (vs His^+ 0.66))

Rationale:
HIS41 makes an electrostatic contact to ASP84 (inter-helix), implying it is positively charged (His^+, propensity 0.66 which is poor). To preserve the electrostatic/salt bridge interaction with ASP84, the replacement must be positively charged. LYS^+ (propensity 0.26) is a strong candidate: it is positively charged like His^+, can form equivalent electrostatic and H-bond contacts to ASP84, and has significantly better helix propensity (0.26 vs 0.66). The pi-alkyl interaction with LEU37 may be partially reduced with Lys vs His, but the helix stability gain is substantial. Best mutation candidate in this helix.

GLN42 | SASA: 74.40 Å² | Intra ONLY | **MUTATE — ALA**

Interaction Types	H-bond + Carbon H-bond
Interaction Details	H-bond donor to SER38 backbone O; carbon H-bond to GLU45 OE1
Current Propensity	Gln: 0.39
Suggested Mutation	Ala (primary); Glu⁰ (secondary) (propensity: 0.00 (primary); 0.16 (secondary))

Rationale:
GLN42 is an Intra-only residue — no inter-helix contacts. Its H-bond donation to SER38 backbone O and C-H bond to GLU45 OE1 are intrahelical and largely backbone-mediated. Mutation to ALA (propensity 0.00) would substantially improve helix propensity. Since the interaction with SER38 backbone O is a backbone contact (not side-chain dependent), the structural impact is minimal. Alternatively, if an H-bond donor is preferred for helix context, Glu⁰ (propensity 0.16) is a secondary option. ALA is the primary recommendation.

GLU45 | SASA: 138.35 Å² | Intra ONLY | **MUTATE — LEU or ALA**

Interaction Types	Conventional H-bond
Interaction Details	H-bond donor to HIS41 backbone O; C-H bond acceptor from GLN42; highest SASA value in the entire helix
Current Propensity	Glu^{-}: 0.40
Suggested Mutation	Leu (primary); Ala (secondary) (propensity: 0.21 (primary); 0.00 (secondary))

Rationale:
GLU45 is the most solvent-exposed residue in the helix (SASA 138.35 Å²) and has Intra-only interactions. Its donation to HIS41 backbone O is a backbone interaction, independent of the Glu side chain charge. No charge-specific contacts exist that would require preserving negative charge. Substitution with LEU (propensity 0.21) or ALA (0.00) would dramatically improve helix propensity while eliminating the partially destabilizing negative charge at this exposed position. LEU is preferred if some side-chain bulk is desired for packing; ALA if maximal helix stability is the goal.

SER46 | SASA: 59.92 Å² | Intra + Inter | **CONDITIONAL**

Interaction Types	H-bond + Carbon H-bond
Interaction Details	Donor to GLN42 and LEU43 backbone O; C-H bond to LEU43 O; inter-helix acceptor from GLY48
Current Propensity	Ser: 0.50
Suggested Mutation	Ala (if backbone-mediated contact confirmed) (propensity: 0.00)

Rationale:
SER46 is at a helix boundary (Intra + Inter) and accepts an inter-helix H-bond from GLY48. Crucially, GLY48 as a donor is a backbone NH — this means the acceptor in SER46 may be the backbone carbonyl rather than the side-chain OH. If confirmed backbone-mediated, mutation to ALA (0.00) or THR is feasible without disrupting the GLY48 contact. If the inter-helix contact involves the Ser side-chain OH, ALA mutation would abolish it. Structural model verification needed before proceeding. Conditionally recommend ALA if backbone-mediated.

Final Recommendations and Priority Ranking

Priority is assigned based on: (a) Intra-only zone, (b) magnitude of propensity improvement, and (c) absence of critical side-chain-dependent contacts.

Priority	Residue	Mutation	Propensity Gain	Justification Basis	Key Rationale
1	**GLU45**	→ LEU or ALA	0.40 → 0.21/0.00	Intra only, highest SASA	Most exposed; no inter-helix contacts; large propensity gain; no charge constraint
2	**HIS41**	→ LYS	0.66 → 0.26	Charge preserved, better propensity	Maintains positive charge for ASP84 electrostatic; major propensity improvement
3	**SER38**	→ ALA	0.50 → 0.00	Intra only, backbone interactions	All contacts backbone-mediated; Ala substitution safe and highly beneficial
4	**GLN42**	→ ALA	0.39 → 0.00	Intra only, H-bond to backbone O	Backbone-mediated donor contact; Ala substantially improves helix propensity
5	**SER46**	→ ALA (conditional)	0.50 → 0.00	Inter-helix from backbone GLY48	Recommended only after confirming inter-helix contact is backbone-mediated

Residues NOT recommended for mutation: SER33 (inter-helix HIS87 H-bond), SER34 (N-cap), LEU37 (optimal propensity + inter-helix pi-alkyl), ARG40 (salt bridge — no better positively-charged substitute). GLN35 requires structural modeling before consideration.